LLM-Based Bloom's Taxonomy Classification Studies – Datasets Used

- **Li et al. (2022)** – *Learning Objectives Dataset*: Collected 21,380 course learning objectives from 5,558 university courses and manually labeled each with one of Bloom's six cognitive levels [1]. This large-scale dataset spans multiple domains. **Status**: *Private* (institution-specific data scraped from course pages; authors indicated plans to open-source the data and code [2] but no public repository is known at present).

- **Sebbaq & El Faddouli (2022)** – *MOOC Learning Objectives*: Created a custom dataset of 2,394 learning objectives from various MOOCs, each annotated with its Bloom's cognitive category [3]. Used to fine-tune BERT for cognitive-level classification of MOOCs. **Status**: *Private* (data was created for the study and not published separately).

- **Waheed et al. (2021)** – *BloomNet Question Set*: Utilized two established question datasets: **(1)** *Yahya et al. (2012) dataset* – 600 open-ended exam questions (100 per Bloom level) from diverse subjects [4]; **(2)** *Haris & Omar (2015) dataset* – 141 open-ended questions compiled from various sources [5]. These were labeled with the original six Bloom's levels (Knowledge, Comprehension, etc.) and used to train/evaluate a transformer-based "BloomNet" classifier. **Status**: *Public* – Both sets have since been made openly available. The 600-question set is published as *Supporting Information* in a PLOS One article [6] (DOI link for dataset), and the 141-question set is also available via that article's supplement [7]. Waheed et al. intended to release all code, data, and configurations upon publication [8].

- **Banujan et al. (2023)** – *Sri Lanka Exam Questions*: Collected a large dataset of 16,584 undergraduate exam questions from multiple Sri Lankan universities, covering various question types (e.g. essays, short answers, MCQs) and subjects. Each question was manually labeled by domain experts according to the **revised** Bloom's taxonomy levels [9]. Used to compare deep learning models (ANN, LSTM with GloVe/BERT) for Bloom's level classification. **Status**: *Private* (dataset described in the paper, but not publicly posted; likely available on request).

- **Gani & Sangodiah (2024)** – *Exam Question Datasets (Combined)*: Compiled multiple Bloom-labeled question banks into a **combined benchmark dataset of ~2,500 questions**. This includes questions from prior works (e.g. the 600 and 141 question sets above, and others) to ensure a balanced distribution of Bloom's categories [10] [11]. **Status**: *Public* – The combined dataset is openly released via Figshare (titled **"Exam Question Datasets"**) [12]. *Access URL*: DOI 10.6084/m9.figshare.22597957 (contains the questions and Bloom level labels, v3 released April 2023).

Other Public Datasets Labeled by Bloom's Taxonomy

- **Yahya et al. (2012) Question Bank** – A widely cited Bloom-labeled dataset of **600 short-answer questions**, evenly distributed across the six cognitive levels [13] [6]. Subjects include sciences, math, technology, etc., making it a general benchmark for question classification. *License*: Public domain via PLOS One (Supporting Info S2 File) [7].

- **Haris & Omar (2015) Question Set** – A smaller corpus of **141 examination questions** (open-ended) labeled by Bloom's levels [5]. These were gathered from diverse online sources and prior studies. Also available through the PLOS One article's supplemental files (S1 File) under a CC-BY license [7].

- **Bloom's Taxonomy Dataset (Devane, 2023)** – A Kaggle dataset assembled for educational analysis of Bloom's levels. It reportedly contains a single CSV of exam questions with their Bloom's level annotations (e.g. questions drawn from computer science and other domains). **Status**: Public on Kaggle. *Access URL*: Available on Kaggle's website (dataset "Blooms Taxonomy Dataset" by Vijay Devane; direct download requires Kaggle login).

- **Revised Bloom's *Quick Flip* Questions** – A reference dataset of prompt questions for each Bloom level. It originates from *Edupress Quick Flip for the Revised Bloom's Taxonomy* (EP-729) and lists prototypical question stems categorized by level (Remember through Create). **Status**: Public domain content (many educators share this list); for example, a PDF from UAMS [14] contains these labeled question prompts. *(Note: This is more of a pedagogical resource than a dataset of assessment items, but it provides clearly labeled examples of questions for each level.)*

- **Additional Research Corpora** – Various studies have developed Bloom-labeled data on specific question types, though not all are fully open. For instance, **Harrison et al. (2019)** combined an existing tagged item bank with 1,500 newly annotated questions (total ~1,500 items) for a classifier of question complexity [15] – but the dataset itself was not publicly released. **Huang et al. (2021)** created a dataset of K-12 *teacher classroom questions* labeled by Bloom levels [16], though this was a limited study and the data is not openly available. Researchers continue to produce labeled sets (often small-scale) in specific domains (e.g. programming exam questions, interview questions), but the **most reusable public datasets** are the ones listed above which cover a range of subjects and come with clear Bloom's taxonomy labels and usage rights.

Sources:

- Li et al., EDM 2022 – automatic Bloom level classification of 21k learning objectives [1].
- Sebbaq & El Faddouli, IRRODL 2022 – BERT on 2,394 MOOC learning objectives [3].
- Waheed et al., arXiv 2021 (BloomNet) – used 600-Q and 141-Q datasets [4] [5]; data to be released [8].
- Banujan et al., IJEDICT 2023 – 16,584-question dataset in Sri Lanka [9].
- Gani & Sangodiah 2024 – combined Bloom dataset on Figshare [12].
- PLOS One 2020 (Mohammed & Omar) – provided Yahya 600-Q and 141-Q data (S1 & S2 Files) [6] [7].

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[1] [2] Automatic Classification of Learning Objectives Based on Bloom's Taxonomy
https://educationaldatamining.org/edm2022/proceedings/2022.EDM-short-papers.55/

[3] (PDF) Fine-tuned BERT Model for Large Scale and Cognitive Classification of MOOCs
https://www.researchgate.net/publication/359973560_Fine-tuned_BERT_Model_for_Large_Scale_and_Cognitive_Classification_of_MOOCs

[4] [5] [8] [2108.07249] BloomNet: A Robust Transformer based model for Bloom's Learning Outcome Classification
https://ar5iv.org/pdf/2108.07249

6 10 11 13 Towards Smarter Assessments: Enhancing Bloom’s Taxonomy Classification with a Bayesian-Optimized Ensemble Model Using Deep Learning and TF-IDF Features
https://www.mdpi.com/2079-9292/14/12/2312

7 Question classification based on Bloom’s taxonomy cognitive domain using modified TF-IDF and word2vec | PLOS One
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0230442

9 files.eric.ed.gov
https://files.eric.ed.gov/fulltext/EJ1413430.pdf

12 Leveraging LLMs for Next Generation Educational Technologies ...
https://arxiv.org/html/2407.17429v2

14 [PDF] Revised Bloom's Taxonomy - UAMS Educational Development
https://educationaldevelopment.uams.edu/wp-content/uploads/sites/57/2025/01/4-Revised-Blooms-and-Objective-Template.pdf

15 Automated Analysis of Learning Outcomes and Exam Questions Based on Bloom’s Taxonomy
https://arxiv.org/html/2511.10903v1

16 Automatic Classroom Question Classification Based on Bloom's ...
https://www.researchgate.net/publication/358464123_Automatic_Classroom_Question_Classification_Based_on_Bloom's_Taxonomy